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A Review on VLSI Floorplanning Optimization using Metaheuristic Algorithms

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***Abstract*—In the VLSI physical design, floorplanning is an essential design step, as it determines the size, shape, and locations of modules in a chip and as such it estimates the total chip area, the interconnects, and, delay. Computationally, VLSI floorplanning is an NP hard problem. So many researchers have suggested various heuristics and metaheuristic algorithms to solve the VLSI floorplan problem. The representation of floorplan is an important aspect of the floorplanning Stage. The floorplan representations have an important impact on the complexity and search space of the floorplan design. In this paper, we included studying and comparing PSO, SA and ACO as optimization algorithms for floorplanning and the representations involved in the VLSI floorplanning problem.**

Keywords—* floorplan; slicing floorplan; non-slicing floorplan; B-tree; Sequence Pair; Genetic Algorithm; Simulated Annealing,Particle Swarm Optimization, Ant Colony Optimization.

I. INTRODUCTION

With the advancement in technology, the design complexity is increasing and the circuit size is getting larger and larger so the physical design becomes very important. Floorplanning is the starting point of VLSI physical design [2] flow. It determines the locations and dimensions of blocks in a chip so as to minimize the chip area and interconnect wirelengths [2]. From computational point of view, it is an NP hard problem [3]. The search space increases exponentially with the increase in blocks therefore finding an optimum solution is a challenging task. The quality of floorplanning depends on its representation [4-15]. The representations of floorplans determine the size of the search space and the complexity of transformation between its representation and its corresponding floorplan. There are many representation methods are such as B*tree, O-tree, Corner Block List (CBL), Bounded Sliced Grid (BSG), Sequence Pair, Transitive Closure Graph (TCG) etc.

II. PROBLEM STATEMENT

VLSI floorplan is to arrange the modules on a chip. The inputs for the floorplanning is a set of m rectangular modules B= {b1,b2,b3………bm}. Widths, heights, and areas of the modules are denoted as wi, hi, and ai respectiviely, 1≤ i≤ m. The objective of floorplaning is to arrange the modules on a chip under the constraints that any two modules are not overlapped and the area, wirelength and other performance indices are optimal. There are two kinds of modules in floorplanning as explained in [2]:

- Hard Modules: A module is called a hard module if its area and aspect ratio are fixed.
- Soft Modules: A module is called a soft module if its area is fixed but aspect ratio can vary.

where, the aspect ratio is defined as the ratio of width to the height of the module. Further the floorplan structure could also be classified in two categories:

- Slicing Floorplan: Slicing structure can be obtained by repetitively cutting the floorplan horizontally or vertically. Fig.1 shows the structure of slicing floorplan.
- Non-Slicing Floorplan: This floorplan can't be cut repetitively. Fig. 2 shows a non slicing floorplan [4].

Floorplaning Cost:

The objectives are to reduce area and total wirelength of interconnects.

$$Cost(F) = \alpha\left(\frac{Area(F)}{Area^*}\right) + (1-\alpha)\left(\frac{Wirelength(F)}{Wirelength^*}\right) \quad (1)$$

where, Area(F) is area of smallest rectangle enclosing all modules. Wirelenth(F) is interconnection cost between modules. Area*, and Wirelength* is estimated minimum area, and minimum wirelength respectively. Where, α is a weight, $0 \le \alpha \le 1$.

The objective of the floorplanning problem is to minimize the cost(F).

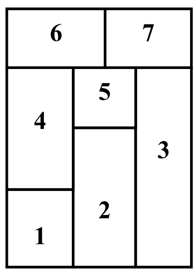

Fig.(1)

Fig.1. Slicing Floorplan

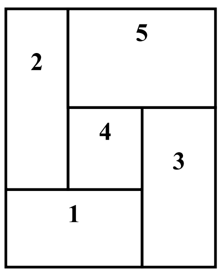

Fig. (2)

Fig. 2. Non-Slicing Floorplan

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III. Reprsentation Schemes For Floorplan

A fundamental problem in VLSI floorplanning is its representation because it determines the size of search space for optimal solution and the complexity of the transformation between a representation and its corresponding floorplan. In this section, we discuss some of the widely used Floorplan Representations in VLSI floorplanning.

A. Sequence Pair

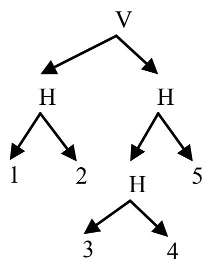
A sequence pair [16] is a representation to form a floorplan. A Sequence Pair consists of an ordered pair of module name. For example, (124536, and 326145) can represent a floorplan of the six modules 1,2,3... 6. To design a Sequence Pair, first creat Constraint Graph Pair i.e. the Horizontal constraint Graph(HCG) and Vertical Constraint Graph(VCG). Given two blocks p and q, in the sequence pair, if p comes before q in T+, i.e., <…p…q…>, and p comes before q in T-, i.e., <…p…q…>, then p is to the left of q. If p comes before q in T+, i.e., <…p…q…>, and p comes after q in T- i.e., <…q…p…>, then p is above q.

B. Polish Notations

Polish Notation is used to model Slicing Floorplans. The Binary Tree is used to record a Polish Expression, E= e1 e2 ...e2n-1 where ei € {1, 2... n, H, V}. Here, each number represents a module and H, V represents a horizontal and vertical cut respectively in the slicing floorplan. The Polish expression is the postfix ordering of a binary tree, which can be obtained from the post-order traversal on a binary tree. It satisfies the balloting property. Polish expression length is 2n-1, where n is number of modules. To revert back a normalized Polish expression E to its corresponding floorplan F, we can use a bottom-up method to recursively combine the slicing sub-floorplans on the basis of E.

C. Bounded Slicing Grid(BSG)

Nakatate et al [5] proposed Bounded slicing Grid for non-slicing floorplan. In BSG, n blocks are placed in special n by n grid. The search space is $O(n!C(n^2,n))$. The search space of BSG is large compare to SP, B*tree, O-tree. Maggie et al [7] proposed a new greedy algorithm on BSG structure, which running in linear time.



12H34H5HV

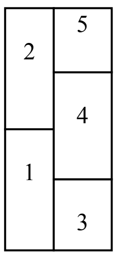


Fig.3(a) *Fig.3(b)*

Fig.3 (a). Binary Tree and its corresponding Polish Notation

Fig.3(b).Transform from polish Notation to its corresponding slicing floorplan

D. B Tree*

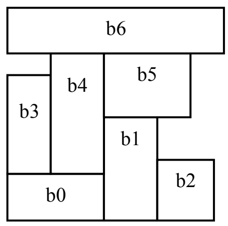
B* Tree [8] is an ordered binary tree and are used to model compacted floorplans [17]. In a compacted floorplan, modules cannot be shifted bottom or left. Fig.4 shows a floorplan and its corresponding B* Tree [11]. The root of B*tree which is denoted by n0 is corresponding to the block b0 on the bottom left corner of the placement. The construction of a B*-tree begins from the root. The left child of the node ni corresponds to the lowest, unvisited block in floorplan. The right child of ni represents the lowest block located above and with its x-coordinate equal to that of bi and its y-coordinate less than that of the top boundary of the module on the left-hand side and adjacent to bi. Each node combination of a B* Tree corresponds to a floorplan, so the solution space consists of all combinations of B*Tree obtained by perturbing the Tree by three movements [11] namely: Node Movement (deleting a node and inserting in some other locations), Node Swap (swapping two blocks) and Block Rotation (No change in tree structure).

E. Transitive Closure Graph(TCG)

The transitive closure of a directed acyclic graph is defined as the Graph G=(V,E). The TCG[9] representation describes the geometric relations among modules based on two graphs, a horizontal transitive closure graph and a vertical transitive closure graph. The TCG[18] consists of two graphs namely horizontal transitive closure graph and vertical transitive closure graph. Horizontal TCG defines horizontal geometric relations between modules and Vertical TCG defines vertical geometric relation between modules. Unlike Sequence Pair, the geometric relations between modules are transparent to TCG as well as its operations. In addition, TCG supports incremental update during operations and keeps the information of boundary modules as well as the shapes and the relative positions of modules in the representation.

To get advantage of SP in TCG at the same time, combination of sequence pair and TCG representation can be used, namely TCG-S.

There are some other floorplan representations reported in literature. Some of them are O-tree, Twin Binary Sequence, Corner Block List (CBL), and TCG-S. Table-I shows a comparative analysis of different floorplan representations, in this table computational complexity, search space and flexibility have been shown.



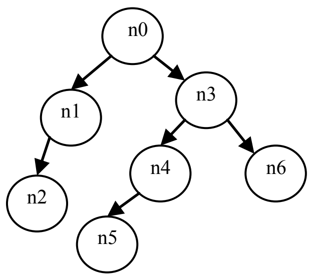


Fig. 4(a) *Fig. 4(b)*

Fig.4. A floorplan and its corresponding b tree*

TABLE I. COMPARISONS AMONG FLOORPLAN REPRESENTATIONS

Floorplan Representations	Computational Complexity	Search Space	Flexibility
Polish	O(n)	$O(n!2^{3n} / n^{1.5})$	Slicing
Corner Block List	O(n)	$O(n!2^{3n})$	Mosaic
Twin Binary Sequence	O(n)	$O(n!2^{3n} / n^{1.5})$	Mosaic
O-tree	O(n)	$O(n!2^{2n} / n^{1.5})$	Compacted
B*-tree	O(n)	$O(n!2^{2n} / n^{1.5})$	Compacted
BSG	$O(n^2)$	$O(n!C(n^2, n)$	General
Transitive Closure Graph	$O(n^2)$	$O(n!)^2$	General
Corner Sequence	O(n)	$O(n!)^2$	General
TCG-S	O(n lg n)	$O(n!)^2$	General

IV. FLOORPLAN ALGORITHMS

As already discussed, floorplanning determines the locations of modules so that goals like minimum area and minimum total interconnect wirelength can be achieved. There are various heuristcs and metaheuristic [19] Algorithms used with different representation methods (like-B*tree, O-tree, CBL [6], Sequence pair etc). These algorithms are Genetic Algorithms, Simulated Annealing, Particle Swarm Optimization[20], Ant Colony Optimization, Differential Evolution Algorithm[21] etc. In next section we will discuss about some of these algorithms.

A. Genetic Algorithm

Genetic algorithm [22] which is part of soft computing used to solve the floorplanning problem such as area and total wirelength optimization on chip. In genetic algorithms [23][24] the total population of chromosomes is firstly considered. Out of these chromosomes which represent the better solution for a given floorplan are considered. The better solution is defined with respect to current population. A fitness function is calculated based on various parameters of floorplan like height of the block, no of modules etc. It can be applied to both slicing and non slicing [25] structures of floorplan. Firstly an initial set of random individuals called an initial population is taken. Then fitness of each individual is evaluated. From the given polish notation we find the number of crossovers. For all these crossovers, mutation is applied, which changes the genetic structure called as genetic mutation. A bit is chosen for a unit in population using the mutation rate. The selected bit is flipped which gives to an entire new unit to be added to new population, fig.5 shows typical GA optimization cycle. The cost is found out for every population in each iteration. If cost is minimum then algorithm stops and gives current population as final optimized result.

B. Simulated Annealing

Simulated Annealing [3] is one of the useful algorithms in each stage of VLSI physical design problem, mainly the floorplanning stage. In this process, firstly a metal is heated at very high temperature, the atoms in the molten metal start moving with respect to each other. As the temperature decreases, atoms slowly get organized and finally form the crystal. The temperature needs to be decreased at very slower rate to get the higher energy state. The main disadvantage of a simulated annealing [26] is that it gives near optimal solution but not stable.

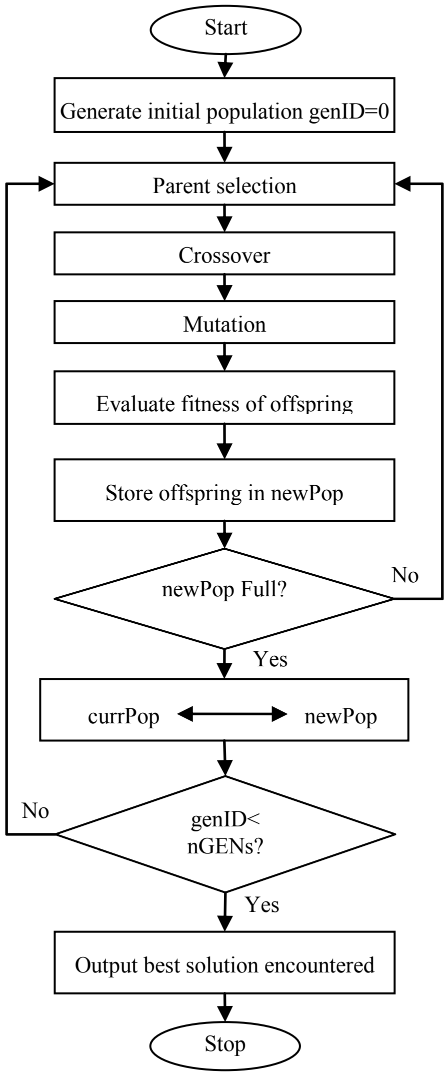


Fig.5. A typical GA optimization cycle

C. Particle Swarm Optimization (PSO)

PSO [27] is an optimization technique inspired by swarm intelligence. PSO [2] is a population-based evolutionary algorithm in which the algorithm is initialized with a population of random solutions. In this algorithm basically learned from animal's activity or behavior to solve optimization problems. Each member of the population is called a particle and the population is called a swarm. Starting with a randomly initialized population and moving in randomly chosen directions, each particle goes through the searching space and remembers the best previous positions of itself and its neighbors. Each particles of swarm communicate its best positions to each other. The next step begins when all particles have been moved. Finally, all particles tend to fly towards better and better positions over the searching process until the swarm move to close to an optimum of the fitness function.

D. Ant Colony Optimization

Ant colony optimization (ACO) [30] is a population based metaheuristic that can be used to find optimal solution to difficult optimization problems. One of the most intensively studied problems in the area of optimization is travelling salesman problem (TSP). The travelling salesman problem is a problem in combinational optimization. It is a NP-hard problem.

The ant-based metaheuristic algorithm consists of three stages, which are the initialization, the construction and the feedback. The primary stage i.e. initialization stage involves the parameters settings such as the number of colonies and the number of ants. The next stage i.e. construction stage involves the construction of path on the basis of pheromone concentration. The last one i.e. the feedback stage deals with the extraction and the reinforcement of the ants travelling experiences obtained during the previous searching path. The ant based metaheuristic algorithm is shown in fig. 6.

- Ant system

The TSP also plays an important role in ACO search: the first ACO algorithm, called Ant System, as well as many of the ACO algorithms proposed subsequently. It was first tested on the TSP. It is an important NP-hard optimization problem that arises in several applications. It is a problem to which ACO algorithms are easily applied. It is easily understandable.

Let there be N cities and M ants. Each ant constructs a tour in greedy fashion and deposits pheromone on the tour it constructs.

τ_{ij} is the pheromone on the link from " i " city to " j "city

$$\tau_{ij}(t+n) \leftarrow (1-\lambda)\tau_{ij}(t) + \Delta\tau_{ij}(t+n) \quad (2)$$

Where, λ = coefficient of evaporation,

$$\Delta\tau_{ij}(t+n) = \sum_{k=1}^{n} \Delta\tau_{ij}^{k} \; \Delta\tau_{ij}(t+n) \quad (3)$$

Now, $\Delta\tau_{ij}^{k} = \begin{cases} 0; if\ ant\ doen't\ travel\ edge\ ij \\ \frac{Q}{L_k}; Otherwise \end{cases}$

Where, L_k is the cost of tour found by ant k and Q is some constant. So we can see that the amount of pheromone that ants deposit on an edge is inversely proportional to length of tour i.e. L_k. Assume, ant at city i moves to city j with probability

$$P \propto \frac{[\tau_{ij}]^{\alpha}[\eta_{ij}]^{\beta}}{\sum[\tau_{ih}]^{\alpha}[\eta_{ih}]^{\beta}} \quad (4)$$

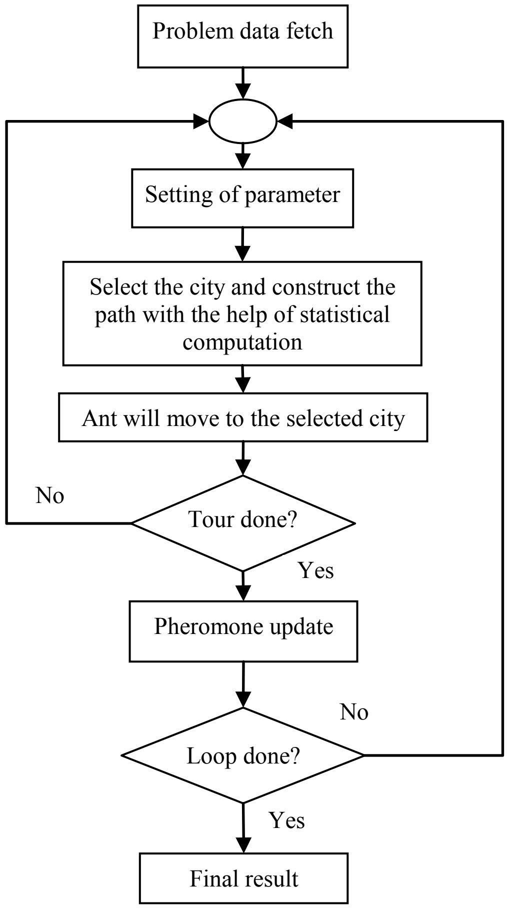


Fig. 6. Ant based meta-heuristic algorithm cycle

Where $[\eta_{ij}]^{\beta}$ is the visibility which is inversely proportional to cost (i,j) and h is the set of allowed cities which means those cities are considered only which will not close the loop prematurely.

V. RESULTS AND DISCUSSIONS

TABLE II. SUMMARY OF RESULT FOR AREA MINIMIZATION IN PUBLISHED DIFFERENT RESEARCH PAPERS

Algorithms	Area(mm2)			
	Apte	Xerox	Ami33	Ami49
SA with B*tree[28]	47.30	20.47	1.36	43.34
SA with sequence pair	48.12	20.69	1.31	38.84
PSO with sequence pair[29]	48.12	20.69	1.31	38.84
PSO with B*tree[17]	46.92	19.55	1.28	41.01
ACO [30]	46.9	19.8	1.20	37.2

TABLE III. THE DETAILS OF MCNC BENCHMARK CIRCUITS

Problems	Benchmark	Modules	Nets	Terminals	Pins
Apte	MCNC	9	97	73	287
Xerox	MCNC	10	203	2	698
Ami33	MCNC	33	123	42	522
Ami49	MCNC	49	408	22	953

We studied different algorithm and different representation of floorplan given by many researchers. From table-II, we found that ACO algorithm with Corner List gives best result in terms of area optimization over other algorithms for different-different MCNC benchmarks given in table-III. After ACO algorithm, PSO algorithm gives good result.

VI. CONCLUSION

In this paper, we have studied about floorplanning problem in the physical design of VLSI Chip. Here floorplanning problem is to arrange and place modules in a chip such a way that goals like minimum area and wirelength could be achieved. There are many floorplan representation have been discussed in this paper. From the analysis, B*-tree and O-tree are most effective in all representations but with less flexibility. As evident from table - II, ACO Algorithm gives best results in area optimization. PSO Algorithm with B*-tree is a close second. We may get better result in future with some new meta-heuristics proposed.

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